Laboratory reports

Rukaiza khan

Pre laboratory requirements

- Cover page with lab title and objectives and measurement variables clearly indicated Supplies and equipment required for the experiment
- Start-up procedure
- Operating procedure
- Shutdown/clean-up procedure
- Emergency shut-down procedure
- General safety hazards and required precautions
- Theoretical analysis and sample calculations: document the theory governing the experiment; and relevant equations used in the calculations, their limitations, and their sources
- Blank data sheets to record experimental results

Front matter

- Front matter includes the title page, executive summary, nomenclature, acronyms and initialisms, and measurement abbreviations.

- Consecutively number the front matter with lowercase Roman numerals (i, ii, iii…) in the footer at the bottom center of the page.
- The title page and executive summary are counted; however, they do not have page numbers.

Title page

- Include on the title page the lab title, course name, completion date, and author names.
- Do not use abbreviations, acronyms, or jargon in the title. The title page is counted as page "i", but it is not numbered.

Executive summary

- It is written after the work is complete in past tense. The executive summary is counted as page “ii” but is not numbered. It conveys the key elements of the lab experiment in concise

  language and includes:
- Background
- Why and how the experiment or test was performed
- The materials and methods used to accomplish the tasks
- Important results, including the extent of agreement between experimental results and theoretical
- predictions, and experimental errors and their estimated effects on the results

Executive summary

- What was discovered, achieved, or concluded
- Relevant recommendation

nomenclature

- List and define all symbols used in the report in alphabetical order, uppercase then lowercase,

Acronyms and initialisms

- List and define acronyms and initialisms used in the report

Measurement abbreviations

- Summarize measurement abbreviations used in the document, including the appendix, by listing the term,
- followed by a comma, and then the abbreviation. Examples include:
- degree Celsius, °C
- minute(s), min
- inch, in.
- liter (no abbreviation)

Report text

- The lab report text includes the Background, Experimental Equipment, and Procedures, Theoretical
- Analysis, Results, Discussion, Conclusions and Recommendations, and References sections.
- Number the report text pages with Arabic numerals in the bottom center of each page (1, 2, 3…).
- information, such as sample calculations, to the appendix. Provide context below each heading for subordinate subheadings.

background

- Provide known information to orient the reader.

Experimental Equipment and Procedures

- If the experimental equipment and procedures are not established, include details of the techniques used.

- Provide enough detail to allow some one of familiar with the general area of investigation to reproduce the experiments from the report information.

- Specify the equipment used, giving credit to the manufacturer. Provide photographs taken during the experiment or a schematic representation of the equipment or simulation program. Include detailed views of unusual or important components and relevant dimensions. Both photographs and schematics must have adequate contrast to be legible when printed in black and white.

- Summarize procedures used (not specific steps). Discuss the type of data collected and the methods used for data collection, reduction, and management.

Theoretical analysis

- Analytically support each experiment by providing mathematical equations used to predict system behavior. Provide and number relevant equations (e.g., Equation 1.3).

- The analysis proceeds from the general (and well-known)basic relationships and evolves to the specific formulae used in the data interpretation. Reference previously derived and readily available analytical
- results using the format in the following example: Equation 6.2 (Smith, 1999).
- Derivation of the referenced equation is not necessary. Present relevant mathematical analyses, in the order required for the experiment, along with supporting explanations and commentaries needed for the reader to understand the
- specific analysis.

results

- Clearly present the experimental results in a condensed, logical manner, using tables or graphs. Point out
- data trends (expound in the Discussion section); and identify concluding observations. Tables and/or
- graphs containing quality-assurance/quality-control data are required.
- Provide raw data and sample calculations in an appendix. Use short statements to present the results without discussion.
- Provide an explanation when needed to prevent incorrect interpretations of the results.

discussion

- Follow a logical progression. Begin with a very brief summary statement of the results and then proceed with a discussion of these results.
- Focus the discussion on the interpretation of the results; note what is "as expected" and what is unexpected. Comment on future investigations if appropriate.
- Include the following, not necessarily in this order:
- 1. Assessment of the reliability of the data and/or calculated results
- 2. Comparison of experimental data to theoretical predictions and the results of similar investigations
- 3. Observed differences and rational explanations for these differences
- 4. Error analyses, noting measurement accuracy and estimated uncertainties
- 5. Unusual and/or unexpected observations, and the most likely causes
- Keep in mind that good paragraph construction presents an idea and then gives supporting details.
- Start sentences with known information and then expand upon this information with new related information.
- Move to the next paragraph when introducing new points.

Conclusion and recommendation

- The conclusions must follow logically and directly from the Results
- It must not include any new information. Make sure the conclusions are supported by data or analysis.
- Direct the reader by stating: "The following conclusions are listed in order of priority and are supported by the results of this study." Then list each conclusion in one or more concise and highly specific (declarative) sentences, using numbers to differentiate each separate conclusion.
- Remember that engineers, managers, and executives are looking for concise statements that clearly tell them what the results and discussion

  have formulated. They are not interested at that point about further investigation or explanation.

  They want the masses of data synthesized into the briefest possible conclusion

- Provide recommendations based on the results and conclusions. Caution the reader about any assumptions and limitations and identify issues that remain unresolved (if appropriate).
- This section is an opportunity to look to the future and imply that the reader has accepted the author's opinions.

references

- Include all references, including established technical specifications (standards) and protocols

appendices

- Include material such as raw data; sample, intermediate, or lengthy calculations;
- long derivatives or detailed information not pertinent to the understanding of the lab report in the appendices.
- The instructor may request inclusion of the Pre Laboratory Report as an appendix.
- Identify all appendix materials within the body of the report. When using multiple appendices, label them with capital letters (A, B, …) and place them in the order referenced in the report.

  Provide a descriptive name for each appendix (e.g., Appendix A: Rheology Data).

  Number each appendix page with the corresponding capital letter followed by Arabic numerals in the bottom center of each page (A1, A2, A3…, B1, B2, B3…).

appendices

- This requires inserting “section breaks/next page” between appendices.